Chapter 4: Postprocessing

Multiple choice questions

Question 1

Level: difficult

Which of the following statements is **FALSE**? Give answer D if all statements are true.

a) The evaluation measures that are used in the single-sample method for evaluating model performance, e.g., the AIC and BIC measures, evaluate the trade-off between the fit of the model and its complexity. They are only meaningful to compare the performance of two models that are built on the same data set.
b) An intuitive interpretation of the AUC performance measure is that it provides an estimate of the probability that a randomly chosen instance of the positive class is correctly ranked higher than a randomly chosen instance of the negative class.
c) For evaluating the performance of a classification model, distance measures such as the divergence measure or Euclidean distance can be used for measuring the separation of the distributions of positives and negatives, taking into account both the difference between the average value of the probability estimates and the variability around the mean of the estimates.
d) All the above statements are true.

Question 2

Level: medium

Which statement about the ROC curve is **TRUE**?

a) The ROC curve is created by plotting the true positive rate (TPR) against the false positive rate (FPR) at various threshold settings, and the ROC space is defined by FPR as y axis and TPR as x axis.
b) The best possible prediction method would yield a point with (x,y)-coordinate equal to (1,0) in the ROC space, representing 100\% sensitivity (no false negatives) and 100\% specificity (no false positives). The (1,0) point is also called a perfect classification.
c) The ROC curve is created by plotting the true positive rate (TPR) against the false positive rate (FPR) at various threshold settings. The true-positive rate is also known as specificity or recall. The false-positive rate is equal to (1 – sensitivity).
d) The ROC curve is created by plotting the true positive rate (TPR) against the false positive rate (FPR) at various threshold settings. The true-positive rate is also known as sensitivity or recall. The false-positive rate is equal to (1 – specificity).

Open questions

Question 1

Explain the differences and similarities between the ROC, Gini and lift curve for evaluating a classification model.

Question 2

How to evaluate a regression model?

Question 3

Why would you adopt an out-of-time evaluation methodology for evaluating a classification model? Explain using the example of a customer churn prediction model.

Question 4

What is the relation between single sample evaluation approaches, complexity and overfitting?

Question 5

What is data leakage and how is it important in evaluating a predictive model?